

QueueLess AI

Intelligent Waiting-Time Prediction & Queue Optimization

Logic League – Ideathon | Galgotias University

1. Abstract

QueueLess AI predicts expected waiting time and recommends the fastest queue using lightweight machine learning.

2. Problem

Queue length alone is unreliable because service speeds and arrival patterns differ.

3. AI/ML

A Random Forest regression model uses six queue-related features. The included data is synthetic for demonstration.

4. Theme

AI/ML → Small / Efficient AI.

5. Prototype Metrics

MAE: 1.85 minutes; RMSE: 2.60 minutes; R^2 : 0.976. These are synthetic-data prototype metrics.

6. Future Scope

Real queue observations, QR check-ins, sensors, mobile app, admin dashboard and continuous retraining.

7. Impact

Potential use in colleges, hospitals, banks, ticket counters and events.